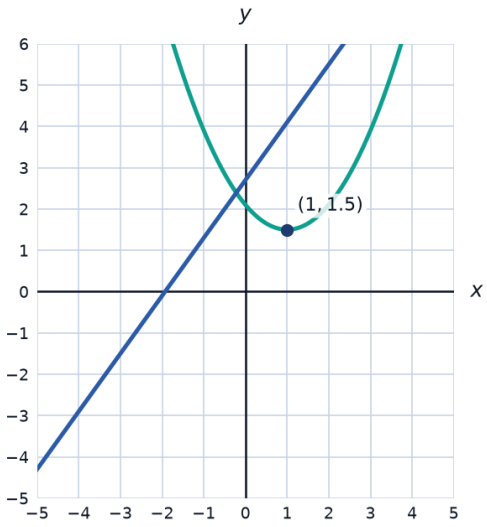
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

[Exponential, Logarithmic & Logistic Dynamics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Vectors, Lines & Parametric Kinematics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.



3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Given that $z_1 = 3 + 4i$ and $z_2 = 1 - 2i$, find $\frac{z_1}{z_2}$ in the form $a + bi$.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given $\ln y = 2\ln x + \ln 3$, express y in terms of x. Hence sketch the graph.

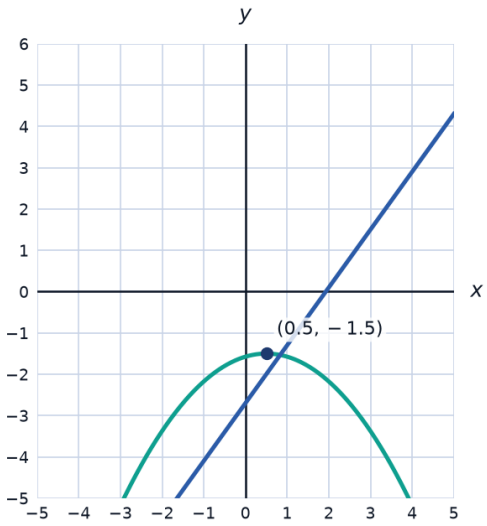
5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the first three non-zero terms of the Maclaurin series for e^{2x}. State the range of validity.

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A bag contains 6 yellow balls, 3 blue balls, and 5 red balls. A ball is drawn at random. Find $P(\text{red})$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A bag contains 5 blue balls, 4 yellow balls, and 8 red balls. A ball is drawn at random. Find $P(\text{red})$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A bag contains 8 blue balls, 6 red balls, and 5 yellow balls. A ball is drawn at random. Find $P(\text{red})$.

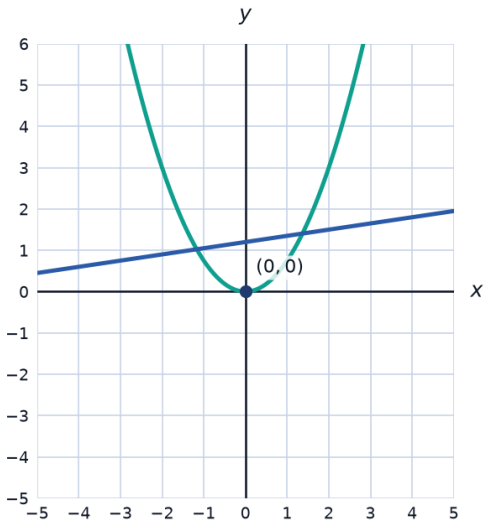
9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A bag contains 4 green balls, 3 blue balls, and 2 red balls. A ball is drawn at random. Find $P(\text{blue})$.

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10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

A bag contains 4 red balls, 2 blue balls, and 2 yellow balls. A ball is drawn at random. Find $P(\text{blue})$.



11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

A bag contains 3 red balls, 3 green balls, and 2 blue balls. A ball is drawn at random. Find $P(\text{blue})$.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

A bag contains 8 blue balls, 4 red balls, and 2 green balls. A ball is drawn at random. Find $P(\text{red})$.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

The data set $1, -1, 3, 2, 6, 10, x$ has mean 5. Find x.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

In a survey, 13 males and 6 females preferred option X, while 15 males and 14 females preferred option Y. Given that a randomly chosen person preferred option X, find the probability that the person is male.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

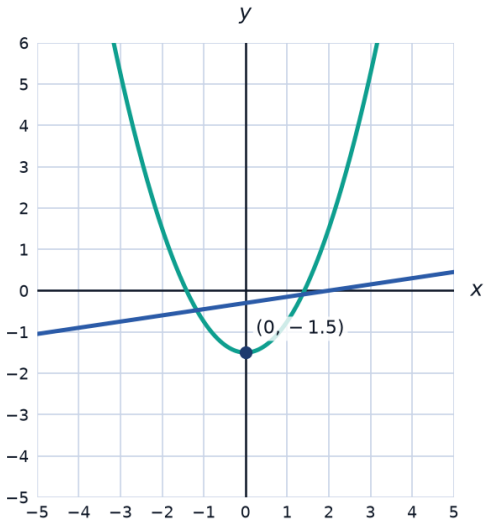
15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 3.01 Graphs & Transformations**

If $P(A) = 0.3$ and $P(B) = 0.5$ and A, B are independent, find $P(A \cup B)$. .

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16 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.01 Coordinate Geometry**

Find the number of distinct permutations of all letters in STATISTICS if all 3 'S's must stay together as a single block.

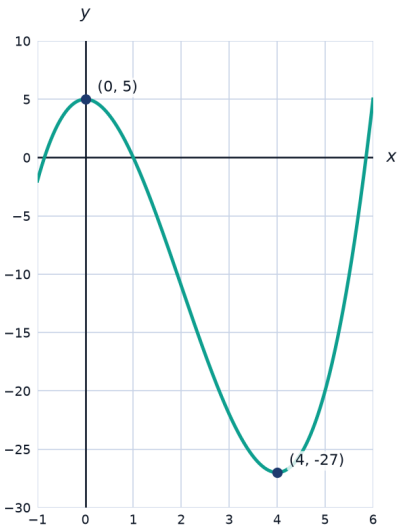


17 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Trigonometry**

A hypothesis test is carried out at the 10% significance level. Find the probability of a Type I error.

18 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Differentiation & Integration**

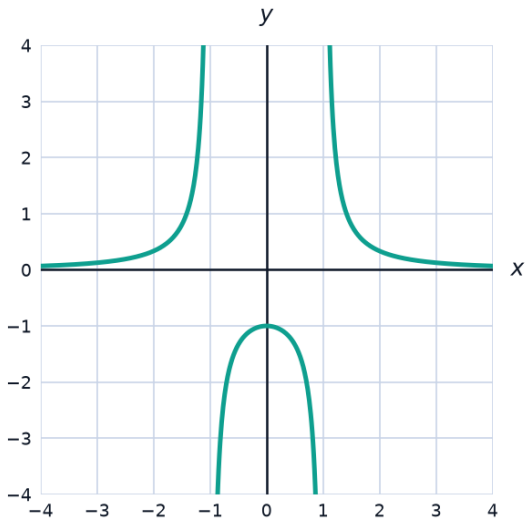
Consider the function $f(x) = x^3 - 6x^2 + 5$. The graph of $y = f(x)$ is shown below. Determine the coordinates of the local maximum of $f(x)$.



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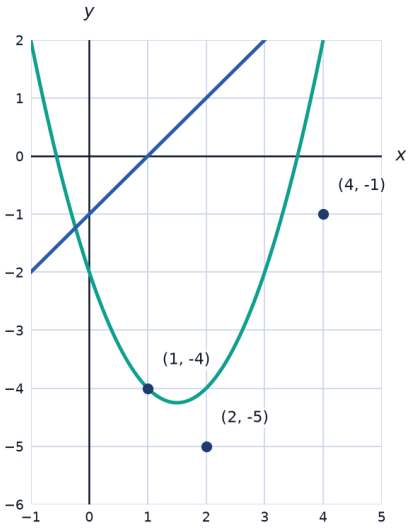
19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 8.01 Functions: Composite & Inverse

Consider the function $f(x) = \frac{1}{x-1}$ and the function $g(x) = x^2$. Let $h(x) = f(g(x))$. The graph of $y = h(x)$ is shown below. Determine the equation of the vertical asymptote(s) of $y = h(x)$.



20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 9.01 Circular Measure

The diagram shows the graph of the function $f(x) = ax^2 + bx + c$. The graph passes through the points $(1, -4)$, $(2, -5)$, and $(4, -1)$. Determine the value of a.



Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The numbers $4, 7, 9, 12, 18$ have a mean of $\bar{x} = 10$. Find the standard deviation σ correct to 2 decimal places.

a. 4.73

b. 5.21

c. 4.12

d. 3.85

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A data set of 20 observations has $\sum x = 240$ and $\sum x^2 = 3280$. Find the variance of the data set.

a. 20

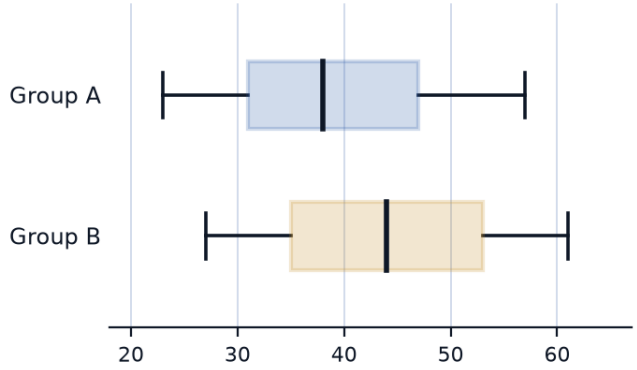
b. 16

c. 24

d. 12

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In a box-and-whisker plot, the lower quartile is $Q_1 = 24$ and the upper quartile is $Q_3 = 42$. According to the $1.5 \times$ IQR rule, what is the boundary for outliers at the upper end?



a. 69

b. 60

c. 65

d. 72

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Two independent events A and B have $P(A) = 0.4$ and $P(B) = 0.5$. Find $P(A \cup B)$.

a. 0.70

b. 0.90

c. 0.20

d. 0.65

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given $P(A) = 0.6$, $P(B) = 0.5$, and $P(A \cap B) = 0.3$, find the conditional probability $P(A \mid B)$.

a. 0.60

b. 0.50

c. 0.30

d. 0.75

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A bag contains 4 red balls and 6 blue balls. Two balls are drawn at random without replacement. What is the probability that both balls are red?

a. $\frac{2}{15}$

b. $\frac{4}{25}$

c. $\frac{1}{6}$

d. $\frac{1}{5}$

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A discrete random variable X has probability function $P(X = x) = kx$ for $x \in \{1, 2, 3, 4\}$. Find the value of k.

a. $\frac{1}{10}$

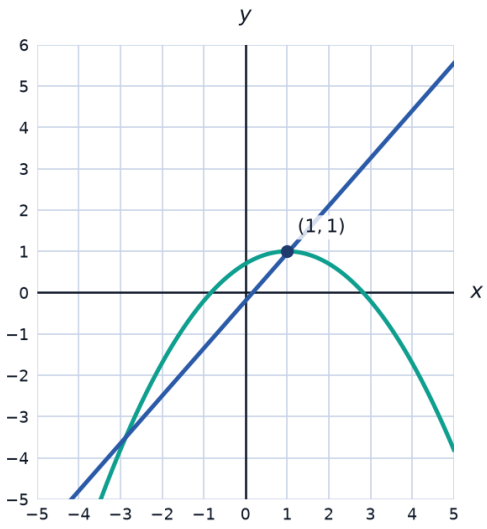
b. $\frac{1}{6}$

c. $\frac{1}{4}$

d. $\frac{1}{12}$

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the expectation $E(X)$ for the distribution where $P(X = 1) = 0.2$, $P(X = 2) = 0.5$, $P(X = 3) = 0.3$.



a. 2.1

b. 2.0

c. 2.3

d. 1.9

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

For a discrete random variable X with $E(X) = 4$ and $E(X^2) = 25$, what is the variance $\mathrm{Var}(X)$?

a. 9

b. 5

c. 16

d. 21

30 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

A random variable $X \sim B(8, 0.25)$. Find $P(X = 2)$ correct to 4 decimal places.

a. 0.3115

b. 0.2076

c. 0.2670

d. 0.1038

31 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

If $X \sim B(100, 0.4)$, find the mean and variance of X.

a. Mean $= 40$, Variance $= 24$

b. Mean $= 40$, Variance $= 16$

c. Mean $= 50$, Variance $= 25$

d. Mean $= 40$, Variance $= 4.9$

32 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

A multiple-choice quiz has 5 questions, each with 4 options and exactly one correct answer. If a student guesses at random on all 5 questions, what is the probability of passing with at least 3 correct answers?

a. $\frac{53}{512} \approx 0.1035$

b. 0.2500

c. 0.0879

d. 0.1563

33 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Group A has 20 data values with mean 15. Group B has 18 data values with mean 10. Find the mean of the combined data set.

a. $\frac{25}{2}$

b. $\frac{240}{19}$

c. 7

d. 8

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A bag contains 7 red and 8 blue balls. A ball is drawn, replaced, then a second ball is drawn. Find $P(\text{both red})$.

a. $\frac{49}{225}$

b. $\frac{112}{225}$

c. $\frac{8}{15}$

d. $\frac{7}{15}$

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A bag contains 4 blue balls, 8 yellow balls, and 3 green balls. A ball is drawn at random. Find $P(\text{yellow})$.

a. $\frac{8}{15}$

b. $\frac{1}{5}$

c. $\frac{7}{15}$

d. $\frac{4}{15}$

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A bag contains 3 red balls, 5 yellow balls, and 8 blue balls. A ball is drawn at random. Find $P(\text{yellow})$.

a. $\frac{11}{16}$

b. $\frac{1}{2}$

c. $\frac{5}{16}$

d. $\frac{3}{16}$

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A bag contains 4 red balls, 3 green balls, and 4 yellow balls. A ball is drawn at random. Find $P(\text{green})$.

a. $\frac{3}{11}$

b. $\frac{2}{11}$

c. $\frac{4}{11}$

d. $\frac{8}{11}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Differentiation & Integration

A bag contains 3 green balls, 7 red balls, and 4 yellow balls. A ball is drawn at random. Find $P(\text{green})$.

a. $\frac{2}{7}$

b. $\frac{1}{2}$

c. $\frac{11}{14}$

d. $\frac{3}{14}$

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 8.01 Functions: Composite & Inverse

A bag contains 4 yellow balls, 6 blue balls, and 3 green balls. A ball is drawn at random. Find $P(\text{green})$.

a. $\frac{6}{13}$

b. $\frac{3}{13}$

c. $\frac{4}{13}$

d. $\frac{10}{13}$

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 9.01 Circular Measure

A bag contains 8 yellow balls, 8 red balls, and 4 blue balls. A ball is drawn at random. Find $P(\text{yellow})$.

a. $\frac{9}{20}$

b. $\frac{2}{5}$

c. $\frac{3}{5}$

d. $\frac{1}{5}$

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Indices, Surds & Quadratics

A bag contains 6 yellow balls, 2 green balls, and 3 blue balls. A ball is drawn at random. Find $P(\text{green})$.

a. $\frac{9}{11}$

b. $\frac{6}{11}$

c. $\frac{3}{11}$

d. $\frac{2}{11}$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 10.01 Series & Binomial Theorem

A bag contains 3 blue balls, 7 green balls, and 2 red balls. A ball is drawn at random. Find $P(\text{green})$.

a. $\frac{1}{4}$

b. $\frac{7}{12}$

c. $\frac{1}{6}$

d. $\frac{5}{12}$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 11.01 Kinematics

A bag contains 6 yellow balls, 4 green balls, and 3 red balls. A ball is drawn at random. Find $P(\text{green})$.

a. $\frac{3}{13}$

b. $\frac{9}{13}$

c. $\frac{4}{13}$

d. $\frac{6}{13}$

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 12.01 Forces & Equilibrium

A bag contains 8 green balls, 7 yellow balls, and 6 blue balls. A ball is drawn at random. Find $P(\text{green})$.

a. $\frac{13}{21}$

b. $\frac{2}{7}$

c. $\frac{1}{3}$

d. $\frac{8}{21}$

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 13.01 Newton's Laws

A bag contains 5 blue balls, 5 green balls, and 2 yellow balls. A ball is drawn at random. Find $P(\text{blue})$.

a. $\frac{7}{12}$

b. $\frac{1}{2}$

c. $\frac{5}{12}$

d. $\frac{1}{6}$

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